

Mahsa Aghazadeh

Machine Learning & Statistical Modelling | Behavioural Signals, Fraud Detection, Large-Scale Event Data

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Daejeon, South Korea; open to relocation

About Me

Data scientist (M.Sc. in Data Science, KAIST, 2025) who turns raw behavioural data into working systems, from typing rhythm and voice to body posture and transactions. Current work focuses on multimodal anomaly detection: learning what is normal for each user from keyboard, stylus, voice and eye-tracking data, and flagging when it changes. The same approach carries over to money-laundering detection.

Education

Korea Advanced Institute of Science & Technology (KAIST) Aug 2023 – Aug 2025
M.Sc. in Data Science, Human Factors and Ergonomics Lab
Thesis: computer vision-based mobile application for ergonomic risk assessment; one journal paper published, a second in preparation

Amirkabir University of Technology Aug 2015 – Aug 2020
B.Sc. in Computer Science

Experience

3R Innovation Daejeon, South Korea
Deputy Team Lead & Data Scientist Dec 2025 – Present

- Predicted ADHD hyperactivity (ROC AUC **0.76**, F2 **0.65**) and inattention (ROC AUC **0.72**, F2 **0.66**) from behavioural data across five attention tasks; compared **9** models (logistic, elastic-net, PLS, tree ensembles, gradient boosting) with nested cross-validation on an imbalanced sample, selecting the model by F2 rather than accuracy.
- Engineered features from keyboard, stylus, voice and eye-tracking recordings through automated pipelines, including speech detection and speaker separation to isolate the child's voice, and hidden Markov models for behavioural states in keystroke time series.
- Built a personal typing-rhythm baseline for each child to flag unusual sessions, such as a different user or a change in the child's state: a per-user generative model over **1.7M** keystrokes from **787** children, scored as a log-likelihood ratio against a population model.
- Analysed, in **103** children across three tasks, which of 51 stylus measures reflect the child rather than the task, using mixed-effects models, ICC and variance decomposition; manuscript in preparation.
- Mentor team members, review analyses, and structure projects from scope to timeline.

KAIST, Human Factors & Ergonomics Lab Daejeon, South Korea
Graduate Researcher, Data Science Aug 2023 – Aug 2025

- Evaluated whether a single smartphone camera is sufficiently accurate for ergonomic risk assessment: designed and conducted a motion-capture study (20 participants, 9 industrial tasks with different levels of occlusion, 3 camera views) benchmarking MediaPipe and MMPose 3D pose estimation against OptiTrack on joint-level accuracy and REBA/RULA score agreement. MediaPipe reached ICC **0.87** (REBA) and **0.92** (RULA), and occlusion produced the largest joint errors.
- Built ErgoYar end to end, a mobile app (Flutter, FastAPI, Supabase) that scores ergonomic risk from video and recommends posture adjustments, and designed its interface; usability with 10 first-time users: SUS **80.5**, uMARS **4.06/5**.

Iranzamin Bank Tehran, Iran
Junior Full-stack Developer, Payments Jan 2021 – Jul 2022

- Built backend services in C# (.NET Core) with SQL Server and REST APIs for core banking systems, including Pichak, Iran's national digital cheque system, translating business and compliance rules into executable decision logic.
- Built web frontends (JavaScript, jQuery, CSS) and mobile UI in Flutter, and developed features in the bank's Java-based business management system.

Selected Projects

Memory-Efficient Money-Laundering Detection
Showed that a fixed-size memory per account is a possible alternative to a full transaction graph for money-laundering detection. On TransXion (3M transactions), XGBoost test average precision rose from **0.237** to **0.364**, close to graph neural networks (approximate comparison). Built an explanation layer for alerts: RAG, an LLM explainer and automated checks. GitHub

Cold-Start Music Recommendation
Recommended new songs with no listening history using contrastive learning: implemented CCFCRec in PyTorch on **1M** user–song interactions. Outperformed content-KNN by **46%** and popularity ranking by **2.4**× (HR@5 **0.77**); ablation attributed the gain to the contrastive objective. GitHub

Efficient Ensemble for Facial Expression Recognition
Approached large-model accuracy at lower cost: an ensemble of three compact CNNs on 35K images came within **1.6 points** of EfficientNet-B3 with **59%** less training time and **44%** fewer MACs; Grad-CAM guided the choice of complementary architectures. GitHub

Skills

ML & Statistics
Anomaly detection, supervised learning (linear models, tree ensembles, gradient boosting), imbalanced classification with cost-sensitive thresholds, sequence and generative models (HMMs), deep and contrastive learning, RAG and LLM output evaluation, feature engineering, nested cross-validation, mixed-effects models, hypothesis testing, experimental design

Programming
Python, C++, C#, Java, Dart, SQL, JavaScript

Libraries
PyTorch, TensorFlow, Scikit-learn, Statsmodels, Pandas, NumPy

Engineering & Tools
FastAPI, Django, .NET Core, SQL Server, Flutter, Git, Figma; familiar with Docker and AWS

Soft Skills
Leadership, ownership, business thinking, communication, fast learning

Languages
English (IELTS 8.0), Korean (TOPIK 5), Persian (Native)

Publications & Conference Presentations

Kweon K, **Aghazadeh M**, Kwon SA, Park S. *Cross-Task Stylus Pressure as a Digital Motor Correlate of Clinician-Rated Hyperactivity in Children.* Manuscript in preparation.

Aghazadeh M, Xiong S. *Computer Vision-Based Monocular 3D Human Pose Estimation and Ergonomic Risk Assessment: The Impact of Model Variants and Camera Views on Kinematic Accuracy and Ergonomic Assessment Validity. International Journal of Industrial Ergonomics*, 116 (2026), 104061. DOI

Aghazadeh M, Xiong S. *ErgoYar: Design and Usability Evaluation of a Mobile Application for Video-Based Automated REBA and RULA Assessment.* In preparation.

Kweon K, **Aghazadeh M**, Kwon SA, Park S, Cho K. *Differential Clinical Prediction of Inattention and Hyperactivity in Children Using Task-Based Digital Phenotyping.* Poster, *Biennial International Congress on Mental Health 2026*, Singapore.

Awards & Leadership

Global Korea Scholarship (GKS), Embassy Track: one of four nationwide recipients in 2022.

Iranian National University Entrance Exam: ranked in the top 2%.

Taekwondo, 4th Dan: assistant coach, team captain and national medalist.